

Exam for Zeyang (Arthur) Yu, “The Mathematical Foundations of Political Methodology”

You have 3 hours for this exam. You may use a calculator and refer to your two class books (Pemberton & Rau and Bertsekas & Tsitsiklis). Please do not use the internet during the exam. Show your work: solutions without work may receive no credit.

1. Given the following hold, what (if anything) do we know about the relationship between sets A and C ?

(a) $A \cap B = A$ and $B \cap C \neq \emptyset$

Nothing.

(b) $A \cup B = A$ and $B \cap C \neq \emptyset$

$A \cap C \neq \emptyset$.

(c) $C \cup D = B$ and $B \cap C = A$

$A = C$.

2. Find the limit of each of the following sequences as $n \rightarrow \infty$, if the limit exists.

(a)

$$\frac{4n^3 - 1}{(n + 1)^3}$$

It's 4.

(b)

$$\frac{(-1)^n n^{\frac{5}{8}}}{7 + n^{\frac{7}{8}}}$$

It's 0.

(c)

$$\frac{(-1)^n n^{\frac{1}{2}}}{n^{\frac{1}{2}} + n^{\frac{2}{5}}}$$

It doesn't exist.

3. Use the (δ, ϵ) definition of a limit to prove that the function

$$f(x) = 1/2x - 10$$

is everywhere continuous.

We need that limit as $x \rightarrow x_o$ of $f(x)$ equals $\frac{1}{2}x_o - 10$ for all x_o . For any $\epsilon > 0$, we can find a $\delta > 0$ such that $|\frac{1}{2}x - 10 - \frac{1}{2}x_o + 10| < \epsilon$ whenever $|x - x_o| < \delta$. Rearranging, we get $\frac{1}{2}|x - x_o| < \epsilon$ when $|x - x_o| < \delta$. Setting $\delta = 2\epsilon$ proves the result.

4. Let

$$f(x) = x^2/(x + 4)$$

- (a) Find the critical points of f and characterize them as maxima, minima, or inflection points.

Critical points at $x = 0, x = -8$; -8 is a local maximum, 0 is a local minimum.

- (b) Find the regions on which f is convex and concave.

f is concave when $x < -4$ and is convex when $x > -4$. f is undefined at $x = -4$.

- (c) Find any asymptotes that f may have.

As $x \rightarrow \pm\infty$ f approaches the line $y = x$. As $x \rightarrow -4$ from above $y \rightarrow \infty$. From below, $y \rightarrow -\infty$.

5. Perform the following matrix operations:

- (a) Find the inverse of

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

It is

$$\begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Find the rank of

$$\begin{bmatrix} 1 & 1 & 5 \\ 0 & 2 & 4 \\ -2 & 0 & -6 \\ 1 & 9 & 21 \end{bmatrix}$$

It is 2.

(c) Check whether the following matrix is positive definite, positive semidefinite, negative definite, negative semidefinite, or none of the above.

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

It is none of the above (the 2X2 principal minors aren't positive for A or non-negative for $-A$).

6. Find and classify all of the critical points of the following functions:

(a)

$$f(x, y) = -10x^2 + 4xy - \frac{2y^2}{5}$$

There are a continuum of critical points: all points of the form $(C, 5C)$ with $C \in \mathbb{R}$. They are all global optima, because f is globally concave: Its Hessian is

$$\begin{bmatrix} -20 & 4 \\ 4 & -\frac{4}{5} \end{bmatrix}$$

so it has determinant 0 and its diagonals are negative. Thus the Hessian is negative semidefinite.

(b)

$$f(x, y, z) = -x^2 + \log(x) - \frac{1}{y} - y - z^2 + z \quad \text{for } x, y, z > 0$$

There is one critical point: $(\frac{1}{\sqrt{2}}, 1, \frac{1}{2})$. It is a global optimum because the function is the sum of concave functions of one variable. No Hessian required.

(c)

$f(x, y) = x^2 + y^2 - xy$ subject to the constraint that $2x - y = 3$

The Lagrangian is

$$L(x, y, \lambda) = x^2 + y^2 - xy - \lambda(2x - y - 3).$$

The first order conditions imply that $2x - y - 2\lambda = 0$, $2y - x + \lambda = 0$, and $2x - y - 3 = 0$. Solving for x and y we get $(\frac{3}{2}, 0)$ as a critical point (incidentally, $\lambda = \frac{3}{2}$). It is a constrained (global) minimum. We know this because $x^2 + y^2 - xy$ is globally convex and our constraint is linear. The Hessian of f is

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

which has determinant $3 > 0$ and positive diagonals.